

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 14/Oct/2025

Time: Three Hours

Max. Marks: 75 Marks

Pharmaceutical Organic Chemistry - III

Q.P. CODE: 5013

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. What is Racemic modification? Discuss the methods of resolution of racemic modification.
OR
Define geometric isomers and explain the methods of nomenclature of geometric isomers.
2. What are heterocyclic compounds? Give their systematic nomenclature and classification.

SHORT ESSAYS

7 x 5 = 35 Marks

3. Define the terms with suitable example a) Diastereoisomers b) Meso compounds
c) Enantiomers
OR
Write a note on elements of symmetry with example.
4. Discuss conformational isomers in Ethane.
OR
Write the synthesis and chemical reactions of Pyrrole.
5. Explain the relative aromaticity and reactivity of Thiophen in contrast to Furan and Pyrrole.
6. Write a note on Fischer-indole synthesis.
7. Outline the Skraups synthesis of Quinoline
8. Explain the mechanism involved in Beckmann's rearrangement.
9. Write the Wolff-Kishner reduction reaction.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Define stereoisomerism with examples.
11. What are chiral molecules? Give example.
12. Mention the name of any two methods of configuration of geometrical isomers.
13. Define Stereospecific reaction with example.
14. What are fused heterocyclic compounds? Give examples.
15. Write the structure and medicinal uses of Furan.
16. Give the reason for basicity of Pyridine.
17. Write the structure and medicinal use of drug containing Azepine nucleus.
18. Give the structure and use of Lithium Aluminium hydride (LiAlH_4).
19. What is Dakin reaction?
